

Data Analytics

Project 3 | Data Cleaning & Preprocessing

PROJECT 3 REPORT

LEVEL - 1 | Data Cleaning

Student	Pratham	Internship	Oasis Infobyte – Data Analytics
Project	Project 3 – Data Cleaning	Level	Level 1
Dataset	AB_NYC_2019.csv (Airbnb NYC)	Tools Used	Python · Pandas · NumPy · Matplotlib · VS Code

Abstract

This project presents a comprehensive data cleaning and preprocessing pipeline applied to the publicly available Airbnb New York City (2019) dataset. Leveraging Python and the Pandas library, the workflow systematically identifies and resolves common data quality issues — including missing values, duplicate records, inconsistent data types, and potential outliers — transforming a raw, messy dataset of 48,895 listings into a clean, analysis-ready form. The report details every step of the cleaning methodology along with quantitative results, code logic, and recommendations for downstream Exploratory Data Analysis (EDA) and machine learning.

1. Introduction

In any data-driven project, the quality of input data directly governs the reliability of the resulting insights and models. Real-world datasets, especially those sourced from web platforms like Airbnb, are rarely perfect — they contain missing entries, typographical inconsistencies, mislabeled categories, and noise. Data cleaning (also referred to as data wrangling or data munging) is the critical first phase of the data science pipeline that addresses these shortcomings.

This project targets the **AB_NYC_2019.csv** dataset — a widely-used open dataset containing information about Airbnb listings in New York City for the year 2019. With nearly 49,000 rows and 16 feature columns spanning host information, geographic coordinates, room types, pricing, and review statistics, the dataset presents several real-world data quality challenges ideal for practicing data cleaning skills.

2. Objectives

1. Load and inspect the raw dataset to understand its structure, dimensions, and data types.
2. Identify all missing (null) values across every column and quantify their extent.
3. Apply appropriate strategies to handle missing data — imputation, placeholder assignment, or retention.

4. Detect and remove duplicate records to ensure each listing appears exactly once.
5. Verify and correct data types so that numerical, categorical, and datetime columns are properly typed.
6. Detect statistical outliers in pricing and availability columns using IQR-based methods.
7. Export a clean, validated dataset ready for EDA, visualization, and machine learning pipelines.

3. Dataset Overview

The dataset used is **AB_NYC_2019.csv**, sourced from Inside Airbnb. It contains **48,895 rows** and **16 columns** capturing diverse attributes of Airbnb listings across New York City's five boroughs.

id	int64	Unique identifier for each listing
name	object	Name/title of the Airbnb listing
host_id	int64	Unique identifier for the host
host_name	object	Name of the host
neighbourhood_group	object	Borough: Manhattan, Brooklyn, Queens, Bronx, Staten Island
neighbourhood	object	Specific neighbourhood within the borough
latitude	float64	Latitude coordinate of the listing
longitude	float64	Longitude coordinate of the listing

room_type	object	Type: Entire home/apt, Private room, Shared room
price	int64	Nightly price in USD
minimum_nights	int64	Minimum number of nights required for booking
number_of_reviews	int64	Total number of reviews received
last_review	object	Date of the most recent review (often missing)
reviews_per_month	float64	Average reviews per month (often missing)
calculated_host_listings_count	int64	Total listings by the host
availability_365	int64	Number of days available per year

4. Tools & Environment

Python 3.x	Core programming language for the entire pipeline.
Pandas	Primary library for data loading, inspection, and transformation.
NumPy	Numerical operations, especially for outlier computation via IQR.
Matplotlib	Visualisation of distributions, missing-value heatmaps, and boxplots.
VS Code	IDE used for script development, debugging, and execution.

5. Methodology

The cleaning pipeline follows a structured, reproducible sequence of eight phases:

1	Phase 1 – Data Loading The dataset is loaded using <code>pd.read_csv('AB_NYC_2019.csv')</code> . Initial inspection with <code>df.shape</code> , <code>df.head()</code> , and <code>df.info()</code> provides an overview of dimensions, column names, non-null counts, and inferred data types.
2	Phase 2 – Structural Inspection Column data types are verified. Columns expected to be numeric (e.g. price, minimum_nights) are confirmed as int64/float64. The last_review column is cast to datetime format for temporal analysis.
3	Phase 3 – Missing Value Detection A missing-value audit is performed using <code>df.isnull().sum()</code> . This reveals four columns with nulls: name (16), host_name (21), last_review (10,052), and reviews_per_month (10,052). A heatmap is plotted with Matplotlib/Seaborn to visualise the spatial distribution of missing data.

4	Phase 4 – Missing Value Handling Each missing column is treated differently based on its semantic meaning. name and host_name receive the placeholder 'Unknown' since they are descriptive fields. reviews_per_month is filled with 0 (no reviews = 0 reviews/month). last_review is intentionally left as NaT, as its absence inherently signals that the listing has never been reviewed — imputing a date would be semantically incorrect.
5	Phase 5 – Duplicate Detection & Removal <code>df.duplicated().sum()</code> identifies any exact duplicate rows. Duplicates (if present) are removed with <code>df.drop_duplicates(inplace=True)</code> , and the row count is verified post-removal.
6	Phase 6 – Data Type Validation All columns are re-inspected. The id, host_id, price, minimum_nights, number_of_reviews, and availability_365 columns are validated as integer types. latitude and longitude are confirmed as float64. Categorical columns (neighbourhood_group, neighbourhood, room_type) are optionally cast to the Pandas Categorical dtype to reduce memory usage.
7	Phase 7 – Outlier Detection The IQR (Interquartile Range) method is applied on the price column. Q1 and Q3 are computed, and the IQR fence ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$) is used to identify extreme outliers. Listings priced at \ \$0 or at extreme highs (\ \$10,000+) are flagged. A boxplot is generated for visual confirmation. Depending on use case, outliers are either removed or retained with a flag column.
8	Phase 8 – Export Cleaned Dataset The final, cleaned DataFrame is exported using <code>df.to_csv('AB_NYC_2019_cleaned.csv', index=False)</code> . A final summary report of null counts and row count is printed for verification.

6. Missing Value Analysis

name	16	0.03%	Fill with 'Unknown'
host_name	21	0.04%	Fill with 'Unknown'
last_review	10,052	20.56%	Retain as NaT (no review = no date)
reviews_per_month	10,052	20.56%	Fill with 0 (no reviews = 0/month)

Note: The high missing rate (20.56%) in last_review and reviews_per_month is directly correlated — listings with no reviews will have both fields empty. This is a domain-specific pattern, not a data error.

7. Key Concepts & Challenges Addressed

Data Integrity

Ensuring every column contains logically consistent values. For example, price values of \ \$0 indicate errors or test entries; minimum_nights values exceeding 365 are suspicious and warranted investigation.

Missing Data Handling

With 10,052 missing entries in two correlated columns, a naive imputation (e.g. mean or median) would introduce false signals. Domain understanding was applied: listings with no reviews genuinely have no last_review date.

Duplicate Removal

Exact duplicate rows may arise from scraping errors or data merges. Removing them ensures that statistical aggregations (averages, counts) are not artificially inflated.

Standardisation

Consistent data types and formats across columns ensure compatibility with downstream tools. Categorical encoding and datetime parsing are key pre-processing steps before modelling.

Outlier Detection

Extremely high price values (e.g. \$10,000/night) or availability of 0 days can skew regression models and summary statistics. IQR-based detection surfaces these cases for analyst review.

8. Results & Cleaned Dataset Summary

Total Rows	48,895	48,895 (no duplicates found)
Total Columns	16	16
Missing: name	16	0 (filled 'Unknown')
Missing: host_name	21	0 (filled 'Unknown')
Missing: reviews_per_month	10,052	0 (filled 0)
Missing: last_review	10,052	10,052 (retained as NaT — intentional)
Duplicate Rows	0	0
Price = \$0 listings	Flagged	Reviewed / removed
Data Types Correct	Partial	Fully validated

9. Conclusion

This project successfully demonstrated a rigorous, end-to-end data cleaning workflow applied to the Airbnb NYC 2019 dataset. Through systematic inspection and targeted remediation strategies, the raw dataset was transformed into a high-quality, analysis-ready resource.

Key takeaways from this project include the importance of **domain knowledge** in making informed imputation decisions (e.g. not imputing last_review), the value of **visualising** missing data patterns before acting on them, and the need for **outlier awareness** even in seemingly clean numerical columns.

The cleaned dataset is now ready for Exploratory Data Analysis (EDA), feature engineering, and machine learning applications such as price prediction models, neighbourhood clustering, and demand forecasting.

10. Future Scope & Recommendations

- Perform full EDA with borough-wise price distributions and room type breakdowns.
- Apply feature engineering: extract month/year from last_review, create price-per-night buckets.
- Build a regression model to predict listing prices based on location, room type, and availability.
- Use NLP techniques on the name column (listing titles) to extract sentiment or keywords.
- Explore clustering algorithms to identify neighbourhood-level Airbnb hotspots.